

An AI-Based Document Translation System with Original Format Preservation

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'n KI-gebaseerde dokumentvertaalsstelsel met oorspronklike formaatbewaring

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Abstract. In the context of the increasing demand for digital document translation, modern translation systems are expected not only to ensure the quality of translated content but also to preserve the original formatting of input documents. However, when translating structurally complex documents such as PDF and DOCX files, many systems still encounter problems such as layout distortion, table misalignment, image displacement, font errors, and limited support for bilingual display modes. This study proposes an artificial intelligence-based document translation system capable of processing two common document formats, DOCX and PDF, while supporting three translation modes: standard translation, side-by-side bilingual translation, and line-by-line bilingual translation. For DOCX documents, the system directly processes document components such as paragraphs, tables, headers, and footers. For PDF documents, the system applies a PDF→DOCX→translation→layout recovery→PDF pipeline to improve format preservation. The system integrates several translation providers, including Google Translate, DeepL Translator, and Gemini 2.5 Flash, to increase flexibility in translation quality and processing options. The experimental evaluation is conducted using two groups of criteria: formatting preservation and translation quality measured by BLEU Mean and BLEU Std. The results show that the proposed system achieves strong format preservation for DOCX documents and promising performance for academic PDF documents. | **Abstrak.** In die konteks van die toenemende vraag na digitale dokumentvertaling, word van moderne vertaalsstelsels verwag om nie net die kwaliteit van vertaalde inhoud te verseker nie, maar ook om die oorspronklike formatering van insetdokumente te bewaar. Wanneer struktureel komplekse dokumente soos PDF- en DOCX-lêers vertaal word, ondervind baie stelsels egter steeds probleme soos uitlegvervorming, tabelwanbelyning, beeldverplasing, fontfoute en beperkte ondersteuning vir tweetalige vertoonmodusse. Hierdie studie stel 'n kunsmatige intelligensie-gebaseerde dokumentvertalingstelsel voor wat in staat is om twee algemene dokumentformate, DOCX en PDF te verwerk, terwyl dit drie vertaalmodusse ondersteun: standaardvertaling, tweetalige vertaling langs mekaar en tweetalige lyn-vir-lyn vertaling. Vir DOCX-dokumente verwerk die stelsel dokumentkomponente soos paragrawe, tabelle, kop- en voettekste direk. Vir PDF-dokumente pas die stelsel 'n PDF→DOCX→vertaling→uitlegherstel→PDF-pyplyn toe om formaatbewaring te verbeter. Die stelsel integreer verskeie vertaalverskaffers,

insluitend Google Translate, DeepL Translator en Gemini 2.5 Flash, om buigsaamheid in vertalingskwaliteit en verwerkingsopsies te verhoog. Die eksperimentele evaluering word uitgevoer deur gebruik te maak van twee groepe kriteria: formatering bewaring en vertaling kwaliteit gemeet deur BLEU Mean en BLEU Std. Die resultate toon dat die voorgestelde stelsel sterk formaatbewaring vir DOCX-dokumente behaal en belowende prestasie vir akademiese PDF-dokumente.

Keywords: Document translation, format preservation, artificial intelligence, PDF, DOCX, bilingual translation, BLEU. | Sleutelwoorde: Dokumentvertaling, formaatbewaring, kunsmatige intelligensie, PDF, DOCX, tweetalige vertaling, BLEU.

1 Introduction

The rapid development of artificial intelligence and large language models has significantly advanced the field of natural language processing, especially machine translation. Modern translation models are no longer limited to converting text from a source language to a target language; they are also capable of handling context, terminology, and writing style more effectively than traditional statistical translation approaches (Bahdanau et al., 2015; Wu et al., 2016; Koehn, 2020). In particular, the Transformer architecture has become a fundamental foundation for many modern

1 Inleiding

Die vinnige ontwikkeling van kunsmatige intelligensie en groot taalmodelle het die veld van natuurlike taalverwerking, veral masjienvertaling, aansienlik bevorder. Moderne vertaalmodelle is nie meer beperk tot die omskakeling van teks van 'n brontaal na 'n doeltaal nie; hulle is ook in staat om konteks, terminologie en skryfstyl meer effektief te hanteer as tradisionele statistiese vertaalbenaderings (Bahdanau et al., 2015; Wu et al., 2016; Koehn, 2020). In die besonder het die Transformer-argitektuur 'n fundamentele grondslag vir baie moderne geword

language processing models due to its ability to capture contextual relationships efficiently in long text sequences (Vaswani et al., 2017). In addition, the development of large language models such as GPT and unified text-to-text models has opened new directions for translation, summarization, and document processing tasks (Brown et al., 2020; Raffel et al., 2020). However, in practical scenarios, users do not only need to translate plain sentences or paragraphs. They often need to translate entire formatted documents such as scientific papers, technical reports, theses, contracts, and administrative forms. These documents are commonly stored in DOCX or PDF formats and contain various formatting components, including headings, paragraphs, tables, images, headers, footers, lists, and page layouts. Therefore, document translation should not be considered only as a content translation problem, but also as a formatting preservation problem.

In many existing document translation systems, preserving the original format remains a considerable challenge. DOCX has a relatively structured representation in which content is organized into paragraphs, tables, styles, headers, and footers, making it possible to process the document directly using programming tools. In contrast, PDF is mainly designed for display and printing, and its content is usually stored based on page coordinates rather than a logical document structure. As a result, when translated text becomes longer or shorter than the original text, problems such as text overflow, table misalignment, overlapping text, and layout changes may occur. Popular translation services such as Google Translate, DeepL, Adobe Acrobat, and AI platforms with file-upload support provide convenient solutions, but they still have limitations in terms of pipeline customization, bilingual presentation, and output format control. Based on these limitations, this study develops an AI-based document translation system that preserves the original document format and supports both DOCX and PDF files with three translation modes. The system uses different processing strategies for different document types: DOCX files are translated directly based on their structured paragraphs and tables, while PDF files are processed using a

taalverwerkingsmodelle vanweë die vermoë daarvan om kontekstuele verhoudings doeltreffend in lang teksreekse vas te vang (Vaswani et al., 2017). Daarbenewens het die ontwikkeling van groot taalmodelle soos GPT en verenigde teks-tot-teks-modelle nuwe rigtings vir vertaal-, opsomming- en dokumentverwerkingstake geopen (Brown et al., 2020; Raffel et al., 2020). In praktiese scenario's hoef gebruikers egter nie net gewone sinne of paragrawe te vertaal nie. Hulle moet dikwels volledige geformateerde dokumente vertaal, soos wetenskaplike referate, tegniese verslae, tesisse, kontrakte en administratiewe vorms. Hierdie dokumente word gewoonlik in DOCX- of PDF-formate gestoor en bevat verskeie formateringskomponente, insluitend opskrifte, paragrawe, tabelle, prente, kopskrifte, voettekste, lyste en bladsyuitelegte. Dokumentvertaling moet dus nie net as 'n inhoudsvertalingsprobleem beskou word nie, maar ook as 'n formateringsbehoudprobleem.

In baie bestaande dokumentvertalingstelsels bly die behoud van die oorspronklike formaat 'n aansienlike uitdaging. DOCX het 'n relatief gestruktureerde voorstelling waarin inhoud in paragrawe, tabelle, style, kop- en voettekste georganiseer word, wat dit moontlik maak om die dokument direk met behulp van programmeringsinstrumente te verwerk. Daarteenoor is PDF hoofsaaklik ontwerp vir vertoon en druk, en die inhoud daarvan word gewoonlik geberg op grond van bladsykoördinate eerder as 'n logiese dokumentstruktuur. As gevolg hiervan, wanneer vertaalde teks langer of korter as die oorspronklike teks word, kan probleme soos teksoorloop, tabelwanbelyning, oorvleuelende teks en uitlegveranderinge voorkom. Gewilde vertaaldienste soos Google Translate, DeepL, Adobe Acrobat en KI-platforms met lêeroplai-ondersteuning bied gerieflike oplossings, maar hulle het steeds beperkings in terme van pyplynaanpassing, tweetalige aanbieding en uitvoerformaatbeheer. Gebaseer op hierdie beperkings, ontwikkel hierdie studie 'n KI-gebaseerde dokumentvertalingstelsel wat die oorspronklike dokumentformaat bewaar en beide DOCX- en PDF-lêers met drie vertaalmodusse ondersteun. Die stelsel gebruik verskillende verwerkingstrategieë vir verskillende dokumenttipes: DOCX-lêers word direk vertaal op grond van hul gestruktureerde paragrawe en tabelle, terwyl PDF-lêers verwerk word deur 'n

PDF→DOCX→translation→layout recovery→PDF pipeline. The proposed system is evaluated through two experimental groups: formatting preservation evaluation and translation quality evaluation, in order to assess both visual structure and linguistic content.

2 Related Work

In the field of document translation, many commercial systems have been developed to help users translate common file formats such as DOCX, PDF, and PPTX. Platforms such as DeepL Document Translation, Google Translate, Google Cloud Translation, and Adobe Acrobat provide online document translation features, allowing users to process multilingual documents for study, research, and professional work. These systems offer advantages in terms of processing speed, translation quality, and usability. However, most commercial platforms are closed systems, meaning that users cannot easily modify the internal processing pipeline, customize format preservation mechanisms, or flexibly control bilingual presentation modes such as side-by-side bilingual translation or line-by-line bilingual translation. For users working with academic or technical documents that contain complex structures,

PDF→DOCX→vertaling→uitlegherstel→PDF-pyplyn te gebruik. Die voorgestelde stelsel word deur twee eksperimentele groepe geëvalueer: formateringsbewaring-evaluering en vertalingskwaliteit-evaluering, ten einde beide visuele struktuur en linguistiese inhoud te assesseer.

2 Verwante werk

Op die gebied van dokumentvertaling is baie kommersiële stelsels ontwikkel om gebruikers te help om algemene lêerformate soos DOCX, PDF en PPTX te vertaal. Platforms soos DeepL Document Translation, Google Translate, Google Cloud Translation en Adobe Acrobat bied aanlyn dokumentvertalingskenmerke, wat gebruikers in staat stel om meertalige dokumente vir studie, navorsing en professionele werk te verwerk. Hierdie stelsels bied voordele in terme van verwerkingspoed, vertaalkwaliteit en bruikbaarheid. Die meeste kommersiële platforms is egter geslote stelsels, wat beteken dat gebruikers nie maklik die interne verwerkingspyplyn kan verander, formaatbewaringsmeganismes kan aanpas of tweetalige aanbiedingsmodusse soos sy-aan-sy tweetalige vertaling of lyn-vir-lyn tweetalige vertaling buigsaam kan beheer nie. Vir gebruikers wat met akademiese of tegniese dokumente werk wat komplekse strukture bevat,

Contribution Title (shortened if too long)

relying only on existing commercial file translation tools may not be sufficient to control formatting errors generated during the translation process.

From an academic perspective, machine translation has evolved significantly from statistical machine translation to neural machine translation and large language models. The attention mechanism in neural machine translation improved the ability to align source and target sentences (Bahdanau et al., 2015), while Google Neural Machine Translation demonstrated the effectiveness of neural networks in large-scale multilingual translation (Wu et al., 2016). Later, the Transformer architecture became a key foundation for modern translation and natural language processing models through its self-attention mechanism (Vaswani et al., 2017). Reviews of neural machine translation also indicate that translation quality has improved considerably, although challenges remain in context handling, domain-specific terminology, and evaluation methodology (Stahlberg, 2020; Koehn, 2020). In addition, large language models such as GPT and unified text-to-text models have expanded the ability to process long text, understand context, and generate more natural outputs (Brown et al., 2020; Raffel et al., 2020).

For document processing, DOCX and PDF formats have very different structural characteristics. DOCX is based on an XML structure, where content is organized into paragraphs, text runs, tables, styles, headers, and footers. This allows a system to iterate through document components, extract and replace text, and preserve most of the original formatting. In contrast, PDF is designed mainly for display and printing, and its text is often stored according to page coordinates. This makes reading-order detection, table recognition, and layout recovery more difficult. Studies on document layout analysis, such as PubLayNet, DocBank, FUNSD, DocVQA, and LayoutLM, show that understanding the visual and structural layout of documents is important for complex document processing tasks (Zhong et al., 2019; Li et al., 2020; Jaume et al., 2019; Mathew et al.,

Bydrae Titel (verkort indien te lank) 3

om slegs op bestaande kommersiële lêervertaalinstrumente te vertrou, is dalk nie voldoende om formateringsfoute wat tydens die vertaalproses gegenereer word, te beheer nie.

Vanuit 'n akademiese perspektief het masjiënvertaling aansienlik ontwikkel van statistiese masjiënvertaling na neurale masjiënvertaling en groot taalmodelle. Die aandagmeganisme in neurale masjiënvertaling het die vermoë verbeter om bron- en teikensinne in lyn te bring (Bahdanau et al., 2015), terwyl Google Neurale Masjiënvertaling die doeltreffendheid van neurale netwerke in grootskaalse veeltalige vertaling gedemonstreer het (Wu et al., 2016). Later het die Transformer-argitektuur 'n sleutelgrondslag vir moderne vertalings- en natuurlike taalverwerkingsmodelle geword deur sy selfaandagmeganisme (Vaswani et al., 2017). Resensies van neurale masjiënvertaling dui ook aan dat vertalingskwaliteit aansienlik verbeter het, alhoewel daar nog uitdagings is in kontekshantering, domeinspesifieke terminologie en evalueringsmetodologie (Stahlberg, 2020; Koehn, 2020). Boonop het groot taalmodelle soos GPT en verenigde teks-tot-teks-modelle die vermoë uitgebrei om lang teks te verwerk, konteks te verstaan en meer natuurlike uitsette te genereer (Brown et al., 2020; Raffel et al., 2020).

Vir dokumentverwerking het DOCX- en PDF-formate baie verskillende strukturele eienskappe. DOCX is gebaseer op 'n XML-struktuur, waar inhoud in paragrawe, tekslopies, tabelle, style, kop- en voettekste georganiseer word. Dit laat 'n stelsel toe om deur dokumentkomponente te herhaal, teks te onttrek en te vervang, en die meeste van die oorspronklike formatering te bewaar. Daarteenoor is PDF hoofsaaklik vir vertoon en druk ontwerp, en die teks word dikwels volgens bladsykoördinate gestoor. Dit maak leesvolgorde opsporing, tabelherkenning en uitlegherwinning moeiliker. Studies oor dokumentuitlegontleding, soos PubLayNet, DocBank, FUNSD, DocVQA en LayoutLM,

2021; Xu et al., 2020). In particular, LayoutLM and LayoutLMv2 combine textual, positional, and visual information to improve document understanding for visually rich documents (Xu et al., 2020; Xu et al., 2021).

A practical approach for PDF processing is to convert PDF files into an editable format such as DOCX, translate the extracted content, and then export the result back to PDF. The PDF→DOCX→translation→PDF pipeline has the advantage of using the flexible editing structure of Word documents, allowing the system to process text by paragraphs, tables, and styles instead of manipulating raw PDF coordinates directly. However, this pipeline also has limitations because the conversion process may introduce intermediate errors such as merged text, fragmented paragraphs, incorrectly detected tables, or modified page layouts. Therefore, for this pipeline to work effectively, preprocessing, DOCX sanitization, layout recovery, and output quality checking are required. For scanned documents or embedded images, OCR is also important for creating a text layer that can be processed, and Tesseract is one of the most widely used OCR engines in both research and practical applications (Smith, 2007).

Machine translation evaluation studies often use automatic metrics such as BLEU, TER, METEOR, and COMET to measure similarity between machine translations and reference translations (Papineni et al., 2002; Snover et al., 2006; Banerjee & Lavie, 2005; Rei et al., 2020). Among them, BLEU is a widely used metric based on n-gram overlap between a system-generated translation and a reference translation

toon dat die begrip van die visuele en strukturele uitleg van dokumente belangrik is vir komplekse dokumentverwerkingstake (Zhong et al., 2019; Li et al., 2020; Jaume et al., 2019; Mathew et al., 2019; Mathew et al., 2020). In die besonder kombineer LayoutLM en LayoutLMv2 tekstuele, posisionele en visuele inligting om dokumentbegrip vir visueel ryk dokumente te verbeter (Xu et al., 2020; Xu et al., 2021).

'n Praktiese benadering vir PDF-verwerking is om PDF-lêers in 'n redigeerbare formaat soos DOCX te omskep, die onttrekte inhoud te vertaal en dan die resultaat terug na PDF uit te voer. Die PDF→DOCX→vertaling→PDF-pyplyn het die voordeel om die buigsame redigeringstruktuur van Word-dokumente te gebruik, wat die stelsel toelaat om teks volgens paragrawe, tabelle en style te verwerk in plaas daarvan om rou PDF-koördinate direk te manipuleer. Hierdie pyplyn het egter ook beperkings omdat die omskakelingsproses intermediêre foute soos saamgevoegde teks, gefragmenteerde paragrawe, verkeerde bespeurde tabelle of gewysigde bladsyuitlegte kan inbring. Daarom, vir hierdie pyplyn om doeltreffend te werk, word voorafverwerking, DOCX-sanering, uitlegherwinning en uitsetkwaliteitskontrolering vereis. Vir geskandeerde dokumente of ingebedde beelde is OCR ook belangrik vir die skep van 'n tekslaag wat verwerk kan word, en Tesseract is een van die mees gebruikte OCR-enjins in beide navorsing en praktiese toepassings (Smith, 2007).

Masjienvertalingsevalueringstudies gebruik dikwels outomatiese maatstawwe soos BLEU, TER, METEOR en COMET om ooreenkomste tussen masjienvertalings en verwysingsvertalings te meet (Papineni et al., 2002; Snover et al., 2006; Banerjee & Lavie, 2005; Rei et al., 2020). Onder hulle is BLEU 'n wyd gebruikte maatstaf gebaseer op n-gram oorvleueling tussen 'n stelselgegenereerde vertaling en 'n verwysingsvertaling

(Papineni et al., 2002). However, BLEU scores must be reported clearly because they can vary depending on tokenization, normalization, and calculation settings (Post, 2018). For document translation with format preservation, translation quality alone is not sufficient. A system may translate content correctly but still fail in practical use if it loses tables, shifts images, distorts layout, or produces an unreadable output document. Therefore, this study evaluates the proposed system from two perspectives: formatting preservation and translation quality.

Based on the above analysis, the research gap lies in combining three factors: translation quality, original format preservation, and flexible bilingual presentation modes. Existing systems usually focus on either content translation or format preservation to a certain extent, but they do not fully support standard translation, side-by-side bilingual translation, and line-by-line bilingual translation within a customizable and extensible pipeline. Furthermore, allowing users to choose among translation providers such as Google Translate, DeepL, and Gemini makes the system more adaptable to different requirements in cost, speed, and translation quality. Therefore, this study proposes an AI-based document translation system that combines DOCX processing, a PDF→DOCX→translation→PDF pipeline, special-content protection, layout recovery, and experimental evaluation of both formatting and translation quality.

3 System Design

The proposed system is designed to solve the problem of translating documents while preserving the original format for two main document types: DOCX and PDF. The overall workflow begins when a user uploads a document through the web interface and selects the target language, translation provider, and translation mode. After receiving the request, the system validates the file format, stores the input document, and creates a background processing job. This asynchronous processing design is suitable for large documents or documents that require multiple intermediate

(Papineni et al., 2002). BLEU-tellings moet egter duidelik gerapporteer word, want dit kan wissel na gelang van tokenisering, normalisering en berekeninginstellings (Post, 2018). Vir dokumentvertaling met formaatbehoud is vertaalkwaliteit alleen nie voldoende nie. 'n Stelsel kan inhoud korrek vertaal, maar steeds misluk in praktiese gebruik as dit tabelle verloor, beelde verskuif, uitleg verdraai of 'n onleesbare uitvoerdokument produseer. Daarom evalueer hierdie studie die voorgestelde stelsel vanuit twee perspektiewe: formatering bewaring en vertaling kwaliteit.

Op grond van bogenoemde ontleding lê die navorsingsgaping in die kombinasie van drie faktore: vertalingskwaliteit, oorspronklike formaatbewaring en buigsame tweetalige aanbiedingswyses. Bestaande stelsels fokus gewoonlik tot 'n sekere mate op óf inhoudvertaling óf formaatbewaring, maar hulle ondersteun nie standaardvertaling, sy-aan-sy tweetalige vertaling en lyn-vir-lyn tweetalige vertaling binne 'n aanpasbare en uitbreidbare pyplyn nie. Verder, deur gebruikers toe te laat om tussen vertaalverskaffers soos Google Translate, DeepL en Gemini te kies, maak die stelsel meer aanpasbaar by verskillende vereistes in koste, spoed en vertaalkwaliteit. Daarom stel hierdie studie 'n KI-gebaseerde dokumentvertalingstelsel voor wat DOCX-verwerking, 'n PDF→DOCX→vertaling→PDF-pyplyn, spesiale-inhoudbeskerming, uitlegherwinning en eksperimentele evaluering van beide formatering en vertaalkwaliteit kombineer.

3 Stelselontwerp

Die voorgestelde stelsel is ontwerp om die probleem van die vertaling van dokumente op te los, terwyl die oorspronklike formaat vir twee hoofdokumenttipes bewaar word: DOCX en PDF. Die algehele werkvloei begin wanneer 'n gebruiker 'n dokument deur die webkoppelvlak oplaai en die teikentaal, vertaalverskaffer en vertaalmodus kies. Nadat die versoek ontvang is, bevestig die stelsel die lêerformaat, stoor die invoerdokument en skep 'n agtergrondverwerkingstaak. Hierdie asinchroniese verwerkingsontwerp is geskik vir groot dokumente of dokumente wat veelvuldige intermediêre

processing steps, especially PDF files. During translation, the user interface can monitor the processing status through a job identifier and allow the user to download the output file when the job is completed.

The system architecture consists of several main components: the user interface, the request-handling API, the background job manager, the file processing service, the translation provider module, the layout recovery module, and the output generation module. The user interface allows users to upload DOCX or PDF files, select the target language, choose a translation provider such as Google Translate, DeepL, or Gemini, and select one of three translation modes: standard translation, side-by-side bilingual translation, and line-by-line bilingual translation. The API validates the uploaded file, stores it, creates a translation record, and initializes a background job. The file processing service then determines the document type and selects the appropriate pipeline. This architecture allows the system to be extended with additional translation providers or new document formats in the future.

verwerkingstappe vereis, veral PDF-lêers. Tydens vertaling kan die gebruikerskoppelvlak die verwerkingstatus monitor deur 'n posidentifiseerder en die gebruiker toelaat om die uitvoerlêer af te laai wanneer die taak voltooi is.

Die stelselargitektuur bestaan uit verskeie hoofkomponente: die gebruikerskoppelvlak, die versoekhantering-API, die agtergrondwerkbestuurder, die lêerverwerkingsdiens, die vertaalverskaffermodule, die uitlegherwinningsmodule en die uitsetgenereringsmodule. Die gebruikerskoppelvlak stel gebruikers in staat om DOCX- of PDF-lêers op te laai, die teikentaal te kies, 'n vertaalverskaffer soos Google Translate, DeepL of Gemini te kies, en een van drie vertaalmodusse te kies: standaardvertaling, tweetalige vertaling langs mekaar en tweetalige lyn-vir-lyn vertaling. Die API bekragtig die opgelaaide lêer, stoor dit, skep 'n vertaalrekord en inisialiseer 'n agtergrondtaak. Die lêerverwerkingsdiens bepaal dan die dokumenttipe en kies die toepaslike pyplyn. Hierdie argitektuur laat toe dat die stelsel in die toekoms uitgebrei kan word met bykomende vertaalverskaffers of nuwe dokumentformate.

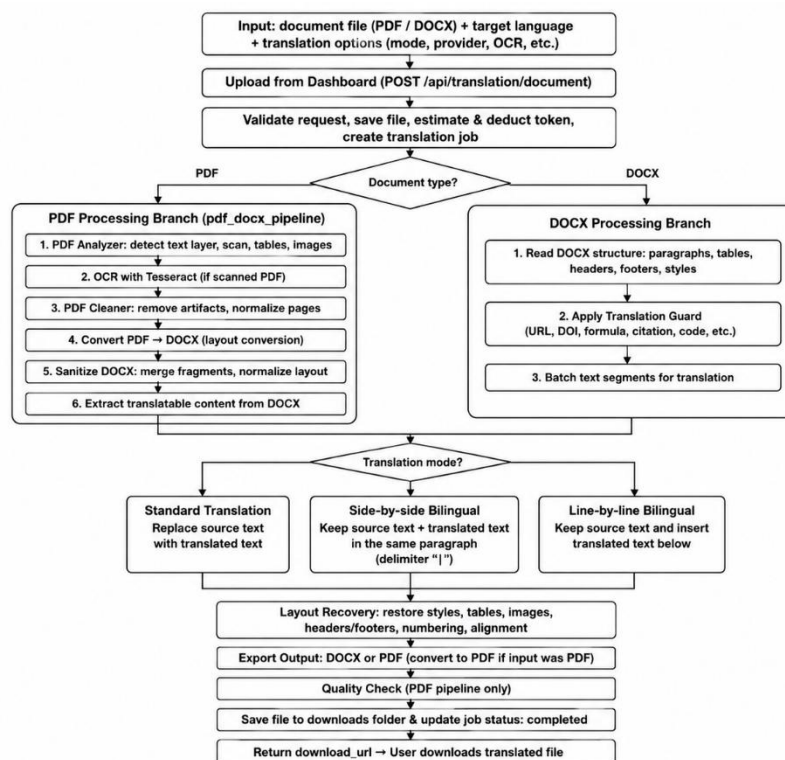


Fig. 1. Workflow of the proposed AI-based document translation system with original format preservation. | *Fig. 1. Werkvloeï van die voorgestelde KI-gebaseerde dokumentvertalingstelsel met oorspronklike formaatbewaring.*

For DOCX documents, the system directly processes the document structure by iterating through paragraphs, tables, headers, and footers. Since DOCX has a clear internal structure, the system can preserve most styles, fonts, margins, and table structures while modifying only textual content. Before sending text to a translation provider, the system applies a Translation Guard mechanism to protect non-translatable elements such as URLs, DOIs, email addresses, citations, formulas, technical symbols, and code snippets. This mechanism is especially important for academic and technical documents because translating such elements incorrectly may change their original meaning. After translation, the translated text is written back into the document according to the selected translation mode.

For PDF documents, the system uses an intermediate conversion pipeline:
PDF→DOCX→translation→layout recovery→PDF.

Vir DOCX-dokumente verwerk die stelsel die dokumentstruktuur direk deur paragrawe, tabelle, kopskrifte en voettekste te herhaal. Aangesien DOCX 'n duidelike interne struktuur het, kan die stelsel die meeste style, lettertipes, kantlyne en tabelstrukture bewaar terwyl slegs tekstuele inhoud gewysig word. Voordat teks aan 'n vertaalverskaffer gestuur word, pas die stelsel 'n Translation Guard-meganisme toe om nie-vertaalbare elemente soos URL's, DOI's, e-posadresse, aanhalings, formules, tegniese simbole en kodebrokkies te beskerm. Hierdie meganisme is veral belangrik vir akademiese en tegniese dokumente omdat die verkeerde vertaling van sulke elemente hul oorspronklike betekenis kan verander. Na vertaling word die vertaalde teks teruggeskryf in die dokument volgens die gekose vertaalmodus.

Vir PDF-dokumente gebruik die stelsel 'n intermediêre omskakelingspylyn:
PDF→DOCX→vertaling→uitlegherstel→PDF.

First, the PDF file is analyzed to determine whether it contains a text layer, whether it is a scanned document, and whether it includes tables, images, or complex page layouts. If the file is scanned or contains image-based text, OCR can be applied to create a processable text layer. The PDF is then cleaned and converted into DOCX. Since PDF-to-DOCX conversion may create issues such as fragmented paragraphs, merged text, incorrect table detection, or abnormal line spacing, the system performs a sanitization step to merge fragmented

Eerstens word die PDF-lêer ontleed om te bepaal of dit 'n tekslaag bevat, of dit 'n geskandeerde dokument is, en of dit tabelle, prente of komplekse bladsyuitlegte insluit. As die lêer geskandeer is of beeldgebaseerde teks bevat, kan OCR toegepas word om 'n verwerkbare tekslaag te skep. Die PDF word dan skoongemaak en in DOCX omgeskakel. Aangesien PDF-na-DOCX-omskakeling probleme kan veroorsaak soos gefragmenteerde paragrawe, saamgevoegde teks, verkeerde tabelbespeuring of abnormale reëlspasiëring, voer die stelsel 'n ontsmettingsstap uit om gefragmenteerde saam te voeg

segments, normalize the layout, and reduce noise before translation. After translation, the Layout Recovery module is used to restore the document layout before exporting the result back to PDF.

The system supports three translation modes to meet different user needs. In standard translation mode, the original text is replaced by the translated text, which is suitable when the user only needs the final document in the target language. In side-by-side bilingual translation mode, the system keeps the original text and places the translated text in the same paragraph, usually separated by the “|” delimiter, allowing users to quickly compare the source and target text. In line-by-line bilingual translation mode, the system keeps the original paragraph and inserts the translated paragraph immediately below it. This mode is useful for language learning, academic reading, and translation checking. Since bilingual modes increase the length of the document, the goal of layout recovery is not to preserve the exact number of pages, but to ensure that the content remains readable, correctly ordered, complete, and free from serious text-overlapping errors.

segmente, normaliseer die uitleg en verminder geraas voor vertaling. Na vertaling word die uitlegherstelmodule gebruik om die dokumentuitleg te herstel voordat die resultaat terug na PDF uitgevoer word.

Die stelsel ondersteun drie vertaalmodusse om aan verskillende gebruikersbehoefte te voldoen. In standaardvertaalmodus word die oorspronklike teks vervang deur die vertaalde teks, wat geskik is wanneer die gebruiker slegs die finale dokument in die doeltaal benodig. In sy-aan-sy tweetalige vertaalmodus hou die stelsel die oorspronklike teks en plaas die vertaalde teks in dieselfde paragraaf, gewoonlik geskei deur die “|” skeidingsteken, wat gebruikers in staat stel om die bron- en teikenteks vinnig te vergelyk. In reël-vir-reël tweetalige vertaalmodus, behou die stelsel die oorspronklike paragraaf en voeg die vertaalde paragraaf onmiddellik daaronder in. Hierdie modus is nuttig vir taalleer, akademiese lees en vertalingskontrole. Aangesien tweetalige modusse die lengte van die dokument vergroot, is die doel van uitlegherwinning nie om die presiese aantal bladsye te bewaar nie, maar om te verseker dat die inhoud leesbaar, korrek georden, volledig en vry van ernstige teksoorvleuelende foute bly.

Algorithm 1

Input:

F: input document (. pdf or . docx)

L: target language

P: translation provider (Google Translate, DeepL, Gemini)

M: translation mode (standard, side-by-side bilingual, line-by-line bilingual)
settings such as OCR and delimiter configuration | Algoritme 1

Opt: optional

Invoer:

F: invoerdokument (. pdf of . docx)

L: doeltaal

P: vertaalverskaffer (Google Translate, DeepL, Gemini)

M: vertaalmodus (standaard, sy-aan-sy tweetalig, reël-vir-reël tweetalig)

Opt: opsionele instellings soos OCR en skeidingskonfigurasie

Output:

F_out: translated document with preserved formatting | Uitset:

F_out: vertaalde dokument met bewaarde formatering

BEGIN

Validate the input document F and target language L | BEGIN

Valideer die invoerdokument F en doeltaal L

IF the file extension of F is not . pdf or . docx THEN Return unsupported
file format error
END IF | AS die lêeruitbreiding van F nie . pdf of . docx DAN Gee
ongesteunde lêerformaatfout terug
EINDE INDIEN

Save F to upload storage
Create a background translation job
Return job_id and status_url to the client | Stoor F om berging op te
laai
Skep 'n agtergrondvertaalwerk
Gee job_id en status_url terug na die kliënt

Set the job status to in_progress | Stel die werkstatus op in_progress

IF F is a PDF document THEN
 Analyze the PDF structure, including text layer, scanned pages, tables, and images | AS F 'n PDF-
dokument DAN is
 Ontleed die PDF-struktuur, insluitend tekslaag, geskandeerde bladsye, tabelle en beelde

IF the PDF is scanned THEN
 Apply OCR to create a searchable PDF IF OCR fails in strict
 mode THEN | AS die PDF DAN geskandeer word
 Pas OCR toe om 'n soekbare PDF te skep AS OCR DAN in
 streng modus misluk

Set the job status to failed
Return error
END IF
END IF | Stel die taakstatus op misluk
Terugkeer fout
EINDE INDIEN
EINDE INDIEN

Clean the PDF document
Convert the PDF document to DOCX Sanitize the
converted DOCX document | Maak die PDF-dokument
skoon
Skakel die PDF-dokument om na DOCX Maak die
omgeskakelde DOCX-dokument skoon

Extract paragraphs, tables, headers, and footers from the DOCX document Apply
translation guard to protect URLs, DOIs, citations, formulas, and special symbols
Translate the extracted content using provider P and mode M
Recover the document layout after translation | Onttrek paragrawe, tabelle, kop- en
voettekste uit die DOCX-dokument Pas vertaalwag toe om URL's, DOI's, aanhalings,
formules en spesiale simbole te beskerm
Vertaal die onttrekte inhoud deur verskaffer P en modus M te gebruik
Herwin die dokumentuitleg na vertaling

IF M is standard translation THEN
Apply the final layout refinement step END IF | AS
M standaardvertaling DAN is
Pas die finale uitlegverfyningstap EINDE IF toe

Convert the translated DOCX document back to PDF Perform
quality checking on the output PDF | Skakel die vertaalde DOCX-
dokument terug na PDF Voer kwaliteitskontrole uit op die uitvoer
PDF

ELSE IF F is a DOCX document THEN
Open the DOCX document directly
Extract paragraphs, tables, headers, and footers
Apply translation guard to protect special content
Translate the DOCX content using provider P and mode M Preserve
the original styles and document structure
END IF | ANDERS AS F DAN 'n DOCX-dokument is
Maak die DOCX-dokument direk oop
Onttrek paragrawe, tabelle, kop- en voettekste
Pas vertaalwag toe om spesiale inhoud te beskerm
Vertaal die DOCX-inhoud deur verskaffer P en modus M te gebruik
Behou die oorspronklike style en dokumentstruktuur
EINDE INDIEN

Save the translated document to download storage Set the job
status to completed
Return the download path of F_out | Stoor die vertaalde
dokument om berging af te laai Stel die taakstatus op voltooi
Keer die aflaai-pad van F_out terug

```
IF any error occurs THEN
  Set the job status to failed with the error message END IF |
INDIEN enige fout DAN voorkom
  Stel die taakstatus op misluk met die foutboodskap END IF
```

```
  The client periodically checks status_url      IF the
job status is completed THEN      Download the
translated document      END IF
END | Die kliënt kontroleer periodiek status_url  AS die
posstatus DAN voltooi is          Laai die
vertaalde dokument af  EINDE INDIEN
EINDE
```

The pseudocode describes the overall processing workflow of the proposed system. For DOCX documents, the system directly processes paragraphs, tables, headers, and

Die pseudokode beskryf die algehele verwerkingswerkvloei van die voorgestelde stelsel. Vir DOCX-dokumente verwerk die stelsel paragrawe, tabelle, opskrifte en

footers to translate textual content while preserving the original formatting. For PDF documents, the system uses DOCX as an intermediate editable format by converting the PDF into DOCX, translating the extracted content, recovering the layout, and exporting the result back to PDF. During translation, the system supports three modes: standard translation, side-by-side bilingual translation, and line-by-line bilingual translation. In addition, special elements such as URLs, DOIs, citations, formulas, and technical symbols are protected to prevent incorrect translation of non-translatable content.

After translation and layout recovery are completed, the system exports the output file according to the original document type. If the input file is DOCX, the system returns a translated DOCX file. If the input file is PDF, the system converts the translated DOCX back to PDF and performs output quality checking. This checking step aims to detect serious errors such as missing content, missing tables, missing images, or layout problems that affect readability. The final file is saved in the download directory, and the download path is returned to the user after the job status changes to completed.

4 Experiments

The experiments were conducted to evaluate the system from two main perspectives: formatting preservation after translation and translation content quality. Separating these two evaluation groups is necessary because document translation depends not only on linguistic quality but also on the ability to preserve the original visual structure of the document. A translation may be semantically correct but still fail in practice if it loses tables, changes the layout, or produces an unreadable output file. Conversely, a document may preserve its formatting well, but the translation may still be insufficient if it is unnatural or uses incorrect terminology.

The experiments were conducted on an Acer Nitro AN515-45 computer running Microsoft Windows 11 Home Single Language 64-bit. The machine uses an AMD Ryzen 55600H with

voettekste om tekstuele inhoud te vertaal terwyl die oorspronklike formatering behoue bly. Vir PDF-dokumente gebruik die stelsel DOCX as 'n intermediêre redigeerbare formaat deur die PDF in DOCX om te skakel, die onttrekte inhoud te vertaal, die uitleg te herstel en die resultaat terug na PDF uit te voer. Tydens vertaling ondersteun die stelsel drie modusse: standaardvertaling, sy-aan-sy tweetalige vertaling en reël-vir-lyn tweetalige vertaling. Daarbenewens word spesiale elemente soos URL's, DOI's, aanhalings, formules en tegniese simbole beskerm om verkeerde vertaling van nie-vertaalbare inhoud te voorkom.

Nadat vertaling en uitlegherwinning voltooi is, voer die stelsel die uitvoerlêer uit volgens die oorspronklike dokumenttipe. As die invoerlêer DOCX is, gee die stelsel 'n vertaalde DOCX-lêer terug. As die invoerlêer PDF is, skakel die stelsel die vertaalde DOCX terug na PDF en voer die kwaliteit kontrolering uit. Hierdie kontrolestep het ten doel om ernstige foute op te spoor soos ontbrekende inhoud, ontbrekende tabelle, ontbrekende beelde of uitlegprobleme wat leesbaarheid beïnvloed. Die finale lêer word in die aflaaigids gestoor, en die aflaaipad word aan die gebruiker teruggestuur nadat die taakstatus na voltooi verander het.

4 Eksperimente

Die eksperimente is uitgevoer om die stelsel vanuit twee hoofperspektiewe te evalueer: formateringbewaring na vertaling en vertalinginhoudkwaliteit. Die skeiding van hierdie twee evalueringsgroepe is nodig omdat dokumentvertaling nie net afhang van linguistiese kwaliteit nie, maar ook van die vermoë om die oorspronklike visuele struktuur van die dokument te bewaar. 'n Vertaling kan semanties korrek wees, maar misluk steeds in die praktyk as dit tabelle verloor, die uitleg verander of 'n onleesbare uitvoerlêer produseer. Omgekeerd kan 'n dokument sy formatering goed bewaar, maar die vertaling kan steeds onvoldoende wees as dit onnatuurlik is of verkeerde terminologie gebruik.

Die eksperimente is uitgevoer op 'n Acer Nitro AN515-45 rekenaar met Microsoft Windows 11 Home Single Language 64-bis. Die masjien gebruik 'n AMD Ryzen 55600H met Radeon Graphics-

Radeon Graphics processor, with 12 logical processors at approximately 3.3 GHz, on an x64-based PC architecture. The system is equipped with 8 GB of RAM and DirectX 12. The development environment uses Python together with the FastAPI framework on the backend for receiving document translation requests, managing background processing jobs, and returning output files to users. The user interface is implemented using HTML, CSS, and JavaScript. For document processing, the system uses pdf2docx for PDF-to-DOCX conversion, DOCX processing libraries for handling paragraphs, tables, headers, and footers, Tesseract OCR for scanned documents or embedded images, and a DOCX-to-PDF conversion tool for generating the final translated PDF output. The translation providers used in the experiments include Google Translate, DeepL Translator, and Gemini 2.5 Flash.

The experimental data consists of DOCX and PDF academic documents containing headings, paragraphs, tables, images, and page layouts. These documents were processed using three translation modes: standard translation, side-by-side bilingual translation, and line-by-line bilingual translation. For format evaluation, the system considers the following criteria: Heading, Text/Style, Table, Layout, and No

verwerker, met 12 logiese verwerkers teen ongeveer 3.3 GHz, op 'n x64-gebaseerde rekenaarargitektuur. Die stelsel is toegerus met 8 GB RAM en DirectX 12. Die ontwikkelingsomgewing gebruik Python saam met die FastAPI-raamwerk op die agterkant vir die ontvangs van dokumentvertalingsversoeke, die bestuur van agtergrondverwerkingstake en die terugstuur van uitvoerlêers aan gebruikers. Die gebruikerskoppelvlak word geïmplementeer met behulp van HTML, CSS en JavaScript. Vir dokumentverwerking gebruik die stelsel pdf2docx vir PDF-na-DOCX-omskakeling, DOCX-verwerkingsbiblioteke vir die hantering van paragrawe, tabelle, kop- en voettekste, Tesseract OCR vir geskandeerde dokumente of ingebedde beelde, en 'n DOCX-na-PDF-omskakelingsinstrument vir die generering van die finale vertaalde PDF-uitset. Die vertaalverskaffers wat in die eksperimente gebruik is, sluit in Google Translate, DeepL Translator en Gemini 2.5 Flash.

Die eksperimentele data bestaan uit DOCX en PDF akademiese dokumente wat opskrifte, paragrawe, tabelle, beelde en bladsyuitelegte bevat. Hierdie dokumente is verwerk met behulp van drie vertaalmodusse: standaardvertaling, tweetalige vertaling langs mekaar en tweetalige lyn-vir-lyn vertaling. Vir formaatevaluering neem die stelsel die volgende kriteria in ag: Opskrif, Teks/Styl, Tabel, Uitleg en Nee

Contribution Title (shortened if too long)

Overflow. Heading measures whether titles and section hierarchy are preserved. Text/Style evaluates whether fonts, paragraphs, and textual presentation remain readable. Table measures whether table structures are preserved. Layout reflects the overall page organization. No Overflow measures whether the output document avoids serious text overflow or overlapping. Minor errors that can be manually corrected, such as slight line shifts or natural page increases caused by longer translated text, are not considered serious errors if they do not affect readability.

The AVG score is calculated as the arithmetic mean of five formatting evaluation criteria: Heading, Text/Style, Table, Layout, and No Overflow. A higher AVG value indicates better formatting preservation and output readability.

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Oorloop. Opskrif meet of titels en afdelinghiërargie behoue bly. Teks/Styl evalueer of lettertipes, paragrawe en teksaanbieding leesbaar bly. Tabel meet of tabelstrukture behoue bly. Uitleg weerspieël die algehele bladsyorganisasie. Geen oorloop meet of die uitvoerdokument ernstige teksoorloop of oorvleueling vermy. Geringe foute wat met die hand reggestel kan word, soos effense lynverskuiwings of natuurlike bladsyverhogings wat veroorsaak word deur langer vertaalde teks, word nie as ernstige foute beskou as dit nie leesbaarheid beïnvloed nie.

Die AVG-telling word bereken as die rekenkundige gemiddelde van vyf formatering-evalueringskriteria: Opskrif, Teks/Styl, Tabel, Uitleg en Geen oorloop. 'n Hoër AVG-waarde dui op beter formateringbewaring en uitvoerleesbaarheid.

Contribution Title (shortened if too long)

relative comparison among machine translation outputs rather than an absolute conclusion about translation quality.

Overall, the experimental results show that the proposed system performs well for DOCX documents and achieves promising results for PDF documents. The difference between DOCX and PDF performance comes from the structural characteristics of each format. DOCX allows direct processing of text and formatting components, while PDF requires several intermediate conversion steps, which increases the possibility of formatting errors. For bilingual translation modes, document length and page count may increase because both the source and translated text are retained. This should not be considered a formatting error as long as the document remains readable, complete, and free from serious text overlapping.

Despite the obtained results, the system still has several limitations that need further improvement. The PDF→DOCX→translation→PDF pipeline cannot guarantee perfect formatting for all types of PDF documents, especially documents with multi-column layouts, complex tables, mathematical formulas, or professional desktop-publishing designs. In addition, scanned PDFs depend heavily on OCR quality; if images are blurred, skewed, noisy, or use unusual fonts, OCR errors may affect translation quality. Since BLEU in this study is used in a cross-output setting due to the absence of a complete manually reviewed reference translation dataset, future work should construct a reference dataset and combine additional metrics such as METEOR, TER, and COMET for a more comprehensive evaluation (Banerjee & Lavie, 2005; Snover et al., 2006; Rei et al., 2020).

5 Conclusion

This study proposed and developed an AI-based

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relatiewe vergelyking tussen masjienvertalingsuitsette eerder as 'n absolute gevolgtrekking oor vertaalkwaliteit.

Oor die algemeen toon die eksperimentele resultate dat die voorgestelde stelsel goed presteer vir DOCX-dokumente en belowende resultate vir PDF-dokumente behaal. Die verskil tussen DOCX- en PDF-prestasie kom van die strukturele kenmerke van elke formaat. DOCX laat direkte verwerking van teks- en formateringskomponente toe, terwyl PDF verskeie intermediêre omskakelingsstappe vereis, wat die moontlikheid van formateringsfoute verhoog. Vir tweetalige vertaalmodusse kan dokumentlengte en bladsytelling toeneem omdat beide die bron- en vertaalde teks behoue bly. Dit moet nie as 'n formateringsfout beskou word nie solank die dokument leesbaar, volledig en vry van ernstige teksoorvleueling bly.

Ten spyte van die resultate wat verkry is, het die stelsel steeds verskeie beperkings wat verdere verbetering benodig. Die PDF→DOCX→vertaling→PDF-pyplyn kan nie perfekte formatering vir alle soorte PDF-dokumente waarborg nie, veral dokumente met multi-kolom-uitlegte, komplekse tabelle, wiskundige formules of professionele desktop-publiserings-ontwerpe. Daarbenewens hang geskandeerde PDF's baie af van OCR-kwaliteit; as beelde vaag, skeef, raserig is of ongewone lettertipes gebruik, kan OCR-foute die vertaalkwaliteit beïnvloed. Aangesien BLEU in hierdie studie in 'n kruisuitsetopstelling gebruik word as gevolg van die afwesigheid van 'n volledige handmatig hersiene verwysingsvertalingsdatastel, behoort toekomstige werk 'n verwysingsdatastel te konstrueer en addisionele maatstawwe soos METEOR, TER en COMET te kombineer vir 'n meer omvattende evaluering (Banerjee & Lavie, 2005; Snover et al. 2006, 20).

5 Gevolgtrekking

Hierdie studie het 'n KI-gebaseerde

document translation system with original format preservation for two main document formats: DOCX and PDF. The system supports three translation modes: standard translation, side-by-side bilingual translation, and line-by-line bilingual translation. It also allows users to select different translation providers, including Google Translate, DeepL Translator, and Gemini 2.5 Flash. For DOCX documents, the system directly processes structured components such as paragraphs, tables, headers, and footers. For PDF documents, the system uses a PDF→DOCX→translation→layout recovery→PDF pipeline to take advantage of DOCX editability and improve format preservation.

The experimental results show that the system achieves very strong format preservation for Word documents and promising results for PDF documents. The bilingual translation modes are useful for helping users compare the original content with the translated content, especially in learning, research, and terminology checking. In terms of translation quality, BLEU Mean and BLEU Std show that the translation providers achieved relatively close results on the test dataset, with Gemini 2.5 Flash obtaining the highest score in the cross-output BLEU experiment. However, to evaluate translation quality more accurately, future research should construct manually reviewed reference translations and combine additional evaluation metrics.

In future work, the system can be improved in several directions. First, PDF processing can be enhanced by improving layout analysis, table recognition, and post-translation layout recovery. Second, OCR quality should be improved for scanned

dokumentvertalingstelsel met oorspronklike formaatbewaring voorgestel en ontwikkel vir twee hoofdokumentformate: DOCX en PDF. Die stelsel ondersteun drie vertaalmodusse: standaardvertaling, sy-aan-sy tweetalige vertaling en reël-vir-lyn tweetalige vertaling. Dit stel gebruikers ook in staat om verskillende vertaalverskaffers te kies, insluitend Google Translate, DeepL Translator en Gemini 2.5 Flash. Vir DOCX-dokumente verwerk die stelsel gestruktureerde komponente soos paragrawe, tabelle, kop- en voettekste direk. Vir PDF-dokumente gebruik die stelsel 'n PDF→DOCX→vertaling→uitlegherstel→PDF-pyplyn om voordeel te trek uit DOCX-redigeerbaarheid en formaatbewaring te verbeter.

Die eksperimentele resultate toon dat die stelsel baie sterk formaatbewaring vir Word-dokumente behaal en belowende resultate vir PDF-dokumente. Die tweetalige vertaalmodusse is nuttig om gebruikers te help om die oorspronklike inhoud met die vertaalde inhoud te vergelyk, veral in leer, navorsing en terminologiekontrolle. Wat vertalingskwaliteit betref, toon BLEU Mean en BLEU Std dat die vertaalverskaffers relatief goeie resultate op die toetsdatastel behaal het, met Gemini 2.5 Flash wat die hoogste telling in die kruisuitset BLEU-eksperiment behaal het. Om vertalingskwaliteit meer akkuraat te evalueer, moet toekomstige navorsing egter handmatig hersiene verwysingsvertalings saamstel en bykomende evalueringsmaatstawwe kombineer.

In toekomstige werk kan die stelsel in verskeie rigtings verbeter word. Eerstens kan PDF-verwerking verbeter word deur uitleganalise, tabelherkenning en na-vertaling uitlegherwinning te verbeter. Tweedens, OCR-kwaliteit moet verbeter word vir geskandeer

PDFs and embedded images. Third, a domain-specific glossary can be added to

ensure terminology consistency in academic and technical documents. Finally,

support for additional formats such as PPTX, XLSX, and HTML can be developed to

broaden the practical applicability of the system.

PDF's en ingebedde beelde. Derdens kan 'n domeinspesifieke woordelys bygevoeg word

verseker terminologiekonsekwentheid in akademiese en tegniese dokumente. Uiteindelik,

ondersteuning vir bykomende formate soos PPTX, XLSX en HTML kan ontwikkel word

die praktiese toepaslikheid van die stelsel te verbreed.

References

1. Bahdanau, D., Cho, K., & Bengio, Y. (2015). *Neural machine translation by jointly learning to align and translate* International Conference on Learning Representations.

<https://arxiv.org/pdf/1409.0473>

2. Banerjee, S., & Lavie, A. (2005). *METEOR: An automatic metric for MT evaluation with improved correlation with human judgments* Proceedings of the ACL Workshop on

Intrinsic and Extrinsic Evaluation] Intrinsieke en Ekstrinsieke Evalueringsmaatreëls vir Masjienvertaling en/of

Summarization, 65–72. <https://aclanthology.org/W05-0909.pdf>

3. Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A.,

Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan,

T., Child, R., Ramesh, A., Ziegler, D. M., Wu, J., Winter, C., Amodei, D. (2020).

Language models are few-shot learners Advances in Neural Information Processing

Systems, 33, 1877–1901. <https://arxiv.org/pdf/2005.14165>

4. Costa-jussà, M. R., Cross, J., Çelebi, O., Elbayad, M., Heafield, K., Heffernan, K.,

Kalbassi, E., Lam, J., Licht, D., Maillard, J., Sun, A., Wang, S., Wenzek, G., Youngblood, A., Akula, B., Barrault, L., Gonzalez, G. M., Hansanti, P., Hoffman, J., Wang, J. (2022).

No language left behind: Scaling human-centered machine translation arXiv preprint

[arXiv: 2207.04672](https://arxiv.org/pdf/2207.04672)

5. Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). *BERT: Pre-training of deep bidirectional transformers for language*

Verwysings

leer om in lyn te bring en te vertaal Internasionale Konferensie oor Leervoorstellings.

verbeterde korrelasie met menslike oordeel Verrigtinge van die ACL-werkswinkel oor

Intrinsieke en Ekstrinsieke Evalueringsmaatreëls vir Masjienvertaling en/of

Opsomming, 65–72. <https://aclanthology.org/W05-0909.pdf>

3. Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A.,

Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan,

T., Child, R., Ramesh, A., Ziegler, D. M., Wu, J., Winter, C., Amodei, D. (2020).

Taalmodelle is min-skoot-leerders Vooruitgang in neurale inligtingverwerking

<https://arxiv.org/pdf/2005.14165>

4. Costa-jussà, M. R., Cross, J., Çelebi, O., Elbayad, M., Heafield, K., Heffernan, K.,

Kalbassi, E., Lam, J., Licht, D., Maillard, J., Sun, A., Wang, S., Wenzek, G., Youngblood, A., Akula, B., Barrault, L., Gonzalez, G. M., Hansanti, P., Hoffman, J., Wang, J. (2022).

Geen taal agtergelaat nie: skaal mensgesentreerde masjienvertaling arXiv-voordruk

[arXiv: 2207.04672](https://arxiv.org/pdf/2207.04672)

tweerigting transformators vir taalverstaan Verrigtinge van

4171–4186. <https://aclanthology.org/N19-1423.pdf>

6. Jaume, G., Ekenel, H. K., & Thiran, J.-P. (2019). *FUNSD: A dataset for form understanding in noisy scanned documents* International Conference on Document

begrip in lawaaierige geskandeerde dokumente Internasionale Konferensie oor Dokument

Analysis and Recognition Workshops. <https://arxiv.org/pdf/1905.13538>

Analise en Erkenning Werkswinkels. <https://arxiv.org/pdf/1905.13538>

7. Junczys-Dowmunt, M., Grundkiewicz, R., Dwojak, T., Hoang, H., Heafield, K.,

7. Junczys-Dowmunt, M., Grundkiewicz, R., Dwojak, T., Hoang, H., Heafield, K.,

Neckermann, T., Seide, F., Hermann, U., Aji, A. F., Bogoychev, N., Martins, A. F. T., &

Neckermann, T., Seide, F., Hermann, U., Aji, A. F., Bogoychev, N., Martins, A. F. T., &

Birch, A. (2018). *Marian: Fast neural machine translation in C++* Proceedings of ACL

Birch, A. (2018). *Marian: Vinnige neurale masjienvertaling in C++* Proceedings of ACL

System Demonstrations, 116–121. <https://aclanthology.org/P18-4020.pdf>

Stelseldemonstrasies, 116–121. <https://aclanthology.org/P18-4020.pdf>

8. Koehn, P. (2020). *Neural machine translation* Cambridge University Press. <https://arxiv.org/pdf/1709.07809>

9. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis,

9. Lewis, P., Perez, E., Pictus, A., Petroni, F., Karpukhin, V., Goyal, N., Kuettler, H., Lewis,

M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). *Retrieval-augmented*

M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). *Herwinning aangevul*

generation for knowledge-intensive NLP tasks Advances in Neural Information

generasie vir kennisintensiewe NLP-take Vooruitgang in neurale inligting

Processing Systems, 33, 9459–9474. <https://arxiv.org/pdf/2005.11401>

10. Li, M., Xu, Y., Cui, L., Huang, S., Wei, F., Li, Z., & Zhou, M. (2020). *DocBank: A benchmark dataset for document layout analysis* Proceedings of the 28th International

maatstafdatastel vir dokumentuitlegontleding Verrigtinge van die 28ste Internasionale

Conference on Computational Linguistics, 949–960. [https://aclanthology.org/2020.coling-](https://aclanthology.org/2020.coling-main.82.pdf)

Konferensie oor rekenaarlinguistiek, 949–960. [https://aclanthology.org/2020.afkoel-](https://aclanthology.org/2020.afkoel-main.82.pdf)

main.82. pdf

hoof.82. pdf

11. Lin, C.-Y. (2004). *ROUGE: A package for automatic evaluation of summaries* Proceedings of the Workshop on Verrigtinge van die werkswinkel oor teksopsomming takke uit, 74–81.

<https://aclanthology.org/W04-1013.pdf>

12. Mathew, M., Karatzas, D., & Jawahar, C. V. (2021). *DocVQA: A dataset for VQA on document images* Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2200–2209.
https://openaccess.thecvf.com/content/WACV2021/papers/Mathew_DocVQA_A_Dataset_for_VQA_on_Document_Imagespaper.pdf
13. Papineni, K., Roukos, S., Ward, T., & Zhu, W.-J. (2002). BLEU: A method for automatic evaluation of machine translation Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics, 311–318.
<https://aclanthology.org/P02-1040.pdf>
14. Post, M. (2018). A call for clarity in reporting BLEU scores Proceedings of the Third Conference on Machine Translation, 186–191. <https://aclanthology.org/W18-6319.pdf>
15. Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W., & Liu, P. J. (2020). Exploring the limits of transfer learning with a unified text-to-text transformer Journal of Machine Learning Research, 21(140), 1–67.
<https://jmlr.org/papers/volume21/20-074/20-074.pdf>
16. Rei, R., Stewart, C., Farinha, A. C., & Lavie, A. (2020). COMET: A neural framework for machine translation evaluation Proceedings of the 2020 Conference on Empirical Methods in
<https://aclanthology.org/2020.emnlp-main.213.pdf>
17. Smith, R. (2007). An overview of the Tesseract OCR engine Proceedings of the International Conference on Document Analysis and Recognition, 629–633.
<https://research.google.com/pubs/archive/33418.pdf>
18. Snover, M., Dorr, B., Schwartz, R., Micciulla, L., & Makhoul, J. (2006). A study of translation edit rate with targeted human annotation Proceedings of the Association for Machine Translation in the Americas, 223–231. <https://aclanthology.org/2006.amta-papers.25.pdf>
19. Stahlberg, F. (2020). Neural machine translation: A review Journal of Artificial Intelligence Research, 69, 343–418. <https://arxiv.org/pdf/1912.02047>
20. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need Advances in Neural Information Processing Systems, 30. <https://arxiv.org/pdf/1706.03762>
21. Wu, Y., Schuster, M., Chen, Z., Le, Q. V., Norouzi, M., Macherey, W., Krikun, M., Cao, Y., Gao, Q., Macherey, K., Klingner, J., Shah, A., Johnson, M., Liu, X., Kaiser, Ł., Gouws, S., Kato, Y., Kudo, T., Kazawa, H., Dean, J. (2016). Google’s neural machine translation system: Bridging the gap between human and machine translation arXiv preprint arXiv: 1609.08144. <https://arxiv.org/pdf/1609.08144>
22. Xu, Y., Li, M., Cui, L., Huang, S., Wei, F., & Zhou, M. (2020). LayoutLM: Pre-training of text and layout for document image understanding Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 1192–1200.
<https://arxiv.org/pdf/1912.13318>
23. Xu, Y., Xu, Y., Lv, T., Cui, L., Wei, F., Wang, G., Lu, Y., Florencio, D., Zhang, C., Che,

- W., Zhang, M., Zhou, L., Huang, Q., & Zhou, M. (2021). LayoutLMv2: Multi-modal pre-training for visually-rich document understanding. *Proceedings of the Annual Meeting of the Association for Computational Linguistics*. <https://aclanthology.org/2021.acl-long.201.pdf>
24. Zhong, X., Tang, J., & Yepes, A. J. (2019). PubLayNet: Largest dataset ever for document layout analysis. *Proceedings of the International Conference on Document Analysis and Recognition*, 1015–1022. <https://arxiv.org/pdf/1908.07836>